The Atree Format: A Scalable Binary-Path Notation for Ancestral Genealogies, with an Application to Lineage Matching

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October 14, 2026

Abstract

We describe `atree`, a compact, human-readable, plain-text notation for *ascending* genealogies (a person's direct ancestors). Each ancestor is named by the path of ancestral sexes from the subject, `M` for father, `F` for mother, so `MF` is the father's mother. This path is exactly the classical Ahnentafel (Sosa–Stradonitz) number written in binary, reading each 0 as father and each 1 as mother. A decimal Ahnentafel number doubles every generation and soon grows unwieldy; an `atree` path instead has one letter per generation, so its length is simply how far back one is looking, and each line stands on its own, needing no surrounding file, so any subset can be shared freely. We give the grammar, exact and reversible conversions to and from Ahnentafel, tool support (GEDCOM import; an HIR/centimorgan calculator), and the limits (ascending trees only; when the same ancestor recurs through cousin marriage, "pedigree collapse", the tree becomes a graph). The main application is *cross-person lineage matching*: when two people's `atree` paths share a tail, the names along it sound alike, and any Y/mtDNA haplogroups agree, that is a measurable, spelling-robust signal that the two trees meet at the same ancestor. About 37 yes/no choices (37 bits) suffice to single out any human who has ever lived, so identifying a person is itself a form of compression.

Keywords: genealogy; Ahnentafel; Sosa–Stradonitz; binary tree; ancestral path; record linkage; phonetic name matching; haplogroup; data compression; Kolmogorov complexity.

1 Introduction

Personal genealogies must be stored and *exchanged* as durable, human- and machine-readable text. The classical Ahnentafel numbering is compact but its decimal identifiers grow exponentially with depth and are opaque as paths. We present `atree`: a sex-path identifier that is the natural coordinate of the ancestor binary tree, fixing both the scalability and the self-containment weaknesses of decimal Ahnentafel.[1] Contributions: (i) the `atree` notation and self-contained plain-text format (the binary path itself is Anderson's [2]; the attribution at the end says what is his); (ii) a formal grammar (with reference converters); (iii) bijective conversions; (iv) a cross-person *lineage-matching* method (combining structural, phonetic, and genetic evidence), with an information-theoretic analysis; and (v) a statement of limitations.

[1]The notation underlying this paper arose earlier, in discussion on the molgen.org genealogy forum and during the author's DNA/genealogy tool work of 2009–2011 (`aisconvert` © 2009 [13]; `aisgedcom/AtreeParser` © 2011 [14]); the binary-tree genealogy format itself is listed by the author as proposed in 2007. He named it "`atree`" and first published the specification as a LiveJournal post on 7 November 2011 [11], adding a Wikipedia section the next day [18]. This paper organizes and extends that material.

2 Background: Ahnentafel / Sosa–Stradonitz

The Ahnentafel system [1, 8] numbers a subject 1, and assigns person n's father $2n$ and mother $2n{+}1$. It is credited to Eytzinger (1590) and takes its common name "Sosa–Stradonitz" from Jerónimo de Sosa (1676) [25] and Stephan Kekulé von Stradonitz (1898) [20], who popularised it. This is the same numbering computer science uses to keep a binary tree in a plain array (a node's two children at positions $2n$ and $2n{+}1$), and listing ancestors in number order simply walks the family tree generation by generation [4, 21]. Writing n in binary, dropping the leading 1, then reading each bit as father (0) or mother (1) gives the genealogical path; `atree` renames those bits `M`/`F`. Anderson [2] set out this reading first, in `f`/`m` form.

3 The `atree` notation

An `atree` identifier is the `M`/`F` path from the subject to an ancestor; the first character encodes the subject's own sex. Formally, $\mathrm{atree}(n) = s\,\mu(b(n))$, where s is the subject's sex letter, $b(n)$ is the binary expansion of n without its leading 1, and μ maps $0 \mapsto$ `M` and $1 \mapsto$ `F`. Each line carries its full path, so any subset of lines is valid and shareable on its own. **No surrogate keys.** The path is a *natural* key: it is derived from the data it names and it states the position it denotes, so an identifier can be read rather than resolved. Neither neighbour has that property. GEDCOM assigns arbitrary surrogate identifiers (`@I1@`) that carry no information and must be dereferenced. Ahnentafel numbers are derived, but opaque: recovering the path from the number takes arithmetic, and the number grows geometrically with depth. `atree` introduces no surrogate identifier at any level.

A single `atree` starts from one subject and lists only that subject's ancestors; a family or a whole database is therefore a *collection* of `atree`s, one per person. Because paths are relative to their subject, the same path string occurs in every tree and denotes a different person in each, so a collection needs one identifier per *tree* to say whose ancestors a block of lines describes. That is the only identifier the format requires. It scales with the number of subjects: N trees need N such labels, not one per ancestor. Joining two collections means keeping the subjects apart, since there are no per-person keys to reconcile, and the matching of Section 10 then works across trees on the paths themselves. Siblings are simply different starting points whose ancestor paths coincide from the parents upward, so their lists share the same tails, the overlap that the matching of Section 10 uses.

4 Grammar

The grammar is BNF with POSIX character classes and regular-expression repetition. One rule is semantic rather than syntactic: a parenthesized group `(...)` is a *date* when its contents parse as a date and it is *not* enclosed by the surname slashes `/.../` (in which case it is a maiden name).

```
<atree>       ::= ( <line> <eol> )*
<line>        ::= <blank> | <comment_line> | <person>
<blank>       ::= <ws>*
<comment_line> ::= <ws>* <comment>
<comment>     ::= "#" [:print:]*

<person>      ::= <path> <haps> <namefield> <datefield>
                  <locfield> <commentfield>
<haps>        ::= ( <ws>+ <haplogroup> )*
<namefield>   ::= ( <ws>+ ","? <ws>* <names> )?
```

```
<datefield>   ::= ( <ws>* <dates> )?
<locfield>    ::= ( <sep> <location> )?
<commentfield> ::= ( <ws>* <comment> )?
<sep>         ::= <ws>+ ","? <ws>* | "," <ws>*

<path>        ::= [MF]+     ; 1st letter = subject's sex;
                            ; rest = father M / mother F
<haplogroup>  ::= "[" ( <hapsys> ":" )? <hapname> "]"
<hapsys>      ::= "Y" | "mt"   ; the two trees share clade letters
<hapname>     ::= [:alnum:] ( [:alnum:] | "-" | "." )*
                            ; stable SNP form, e.g. R-M269, N-M178
<names>       ::= <name_text>? ( <ws>* <surname>
                  ( <ws>* <name_text> )? )?
<name_text>   ::= <word> ( <ws>+ <word> )*
<word>        ::= <name_char>+
<name_char>   ::= [:alnum:] | "'" | "-" | "." | "?"
<surname>     ::= "/" <name_text>? <maiden>? "/"
<maiden>      ::= "(" <name_text>? ")"

<dates>       ::= "(" <ws>* <date_or_range> <ws>* ")"
<date_or_range> ::= <gdate> ( <ws>* "-" <ws>* <gdate> )?
<gdate>       ::= <approx>? ( <full_date> | <short_date>
                  | <year> ) | "?"
<approx>      ::= ( "ABT" | "BEF" | "AFT" | "EST" ) <ws>+ | "~"
<full_date>   ::= <day> <ws>+ <month> <ws>+ <year>
<short_date>  ::= <month> <ws>+ <year>
<day>         ::= [:digit:]{1,2}
<year>        ::= [:digit:]{1,4}
<month>       ::= "JAN"|"FEB"|"MAR"|"APR"|"MAY"|"JUN"
                | "JUL"|"AUG"|"SEP"|"OCT"|"NOV"|"DEC"

<location>    ::= <loc_char>+
<loc_char>    ::= [:print:] except "#"
<ws>          ::= [:blank:]
<eol>         ::= "\n"
```

5 Line syntax and examples

Names precede an optional surname in `/.../` (GEDCOM style) with a maiden name in parentheses; `//` and `?` mark unknowns; dates use `ABT`/`BEF`/`AFT`/`EST` and ranges. A worked ascending tree (excerpt); each person is one line:

```
M    [Y:N1c1] ...
MM   [Y:N1c1] ... /Gavrilov/
MF   [mt:H10a1] Zoya Vasilyevna /Gavrilova (Guseva)/ (20 SEP 1941 - 20 OCT 1984)
MFM  Vasiliy Nikolayevich /Gusev/ (~1914 - 1963)
MFF  [mt:H10a1] Anastasiya Ivanovna /Guseva (Marinova)/ (20 DEC 1913 - 03 JUN 2006)
MFMF Evdokiya Nikolayevna /Guseva (?)/ (?), Milna, daughter of merchant
```

6 Conversions

Proposition 1. *For a fixed subject sex,* atree(·) *is a bijection between Ahnentafel numbers* $n \geq 1$ *and* `atree` *paths.*

Proof. Writing n in binary and dropping the leading 1 is a bijection from $\{n \geq 1\}$ onto $\{0,1\}^*$; the relabelling $0 \mapsto$ `M`, $1 \mapsto$ `F` is a bijection onto $\{\texttt{M},\texttt{F}\}^*$; and the subject-sex letter is prepended (and, in reverse, normalised) invertibly. The two converters below realise both directions. □

The reference converters are two short shell scripts, `atree2s.sh` (Ahnentafel → `atree`) and `s2atree.sh` (`atree` → Ahnentafel), reproduced in Appendix B; they are preferred over the Java toolkit for portability, e.g. `s2atree.sh MFFMMMMFFFMMF` yields `7225`.

Person	Ahnentafel	Binary	`atree` (subj. F)	`atree` (subj. M)
Self	1	1	F	M
Father	2	10	FM	MM
Mother	3	11	FF	MF
Paternal GF	4	100	FMM	MMM
Paternal GM	5	101	FMF	MMF
Maternal GF	6	110	FFM	MFM
Maternal GM	7	111	FFF	MFF

7 Properties

Record independence. One line is one person, and it carries its full path, so it means the same thing alone as in the file it came from. This is the property the rest follow from. Because no line refers to another, a line can be questioned, corrected, re-sorted, dropped or sent on its own, which matches how genealogical research actually proceeds: one ancestor at a time, provisionally, and revised years later. The grammar supports the provisional state directly, with `?` and `//` for unknown fields and `(?)` for an unknown maiden name (Appendix A), so partial knowledge is written down. **Hand-repairability.** Having no cross-references to keep consistent, the file can be edited in any text editor without a parser, and a damaged file degrades line by line rather than as a whole. **Tool-independence.** A format is durable to the extent that it can be read and repaired without its software, which is the practical content of the digital-preservation argument for plain text. This work supplies its own evidence. The 2009–2011 Java tooling described in Section 8 no longer builds unmodified; the files it produced remain readable and editable, and the reference converters of Appendix B still run. **Short identifiers.** A path has one letter per generation, so its length is just the generation distance (about $\log_2$ of the Ahnentafel number), and there is no large number to overflow, unlike a decimal Ahnentafel. **Natural tree order.** The letters give an ancestor's generation and position directly, so paths sort into family-tree order, and being UTF-8 and one record per line the format is easy to compare (diff) and merge (version-control friendly).

8 Tooling and interoperability

`aisgedcom` [14] is the `atree` reference implementation: a parser (`AtreeParser`), GEDCOM↔`atree` conversion, the canonical grammar and example records, and a two-relatives relationship calculator. `aisconvert` [13] (personal genomics toolkit) provides the HIR/Half-IBD calculator, which measures shared autosomal segments and reports centimorgans across autosomes 1–22. It computes sharing; it does not infer relationships. The X chromosome is reported separately, since its sex-dependent transmission traverses only a restricted subset of pedigree paths, so X centimorgans must not be pooled into a total from which relationship distance is later inferred.

The `atree` term appears in third-party genealogy discussion, for example a Gramps forum thread proposing a "Dual Atree chart for cousins" [5]. The wording there follows the Wikipedia section the author added in 2011 [18], so the term's propagation is traceable to that edit and is not independent adoption.

9 Limitations and open problems

`atree` covers the *ascending* (two-parent) tree; descendant trees branch many ways and are out of scope. **Pedigree collapse** (the same ancestor appearing more than once through cousin marriage) gives more than one path to a single real person. A path is unique as a *position*, not as a *person*. The full pedigree is then a graph, one ancestor reached by several paths, and storing it compactly becomes a matter of sharing the repeated sub-trees (a compression problem). A full genealogy, covering all branches or a whole database, is therefore kept as a *collection* of `atree`s, one ascending tree per subject (Section 3). This covers the whole graph at the cost of repeating shared ancestors across trees, the repetition that a collapse-aware, sub-tree-sharing encoding would remove.

10 Application: cross-person lineage matching

The central application is matching people who share ancestors, a long-standing problem in genealogy: deciding when two records mean the same person. When persons A and B share a common ancestor X, every ancestor *above* X is the same in both trees; so although their paths *to* X differ (they start from different people), the `M`/`F` steps from X upward, the tail of each path, are identical, and the names along that tail agree apart from spelling. Four signals combine. The first two need only the written record: (i) a shared **tail** of the path (the longer the shared tail, the more generations in common, and this uses no names at all); and (ii) **name sound-alike** agreement [22, 24] to absorb spelling differences. Phonetic coding is *optional* and is performed by the matching software at comparison time, not required by the format: names are stored as written, in whatever spelling is convenient to the user, and the matcher applies Soundex/Metaphone (or any chosen scheme) when scoring. All normalisation is deferred to the software, keeping the stored file human-convenient. A missing or extra generation shifts the tail by a position or two, so the tails are compared by approximate alignment, allowing an occasional inserted or dropped step, rather than by exact match. Writing each path deep-ancestor-first and grouping paths by their shared opening avoids comparing every pair: records with the same deep ancestry fall into the same group, so a match becomes a lookup. In the language of record linkage, the shared tail is a new grouping key that does not depend on names. The two extreme paths are exactly the DNA-trackable lines: all-`M` (`MMM`…) is the patrilineal (Y-DNA) line, all-`F` (`FFF`…) the matrilineal (mtDNA) line, and the haplogroup field is the third signal, confirming or refuting a match along those two edges alone. A fourth signal is independent of the written record altogether. Autosomal half-identical regions (HIR), measured in centimorgans, are shared in an amount that falls with relationship distance, and the `aisconvert` calculator [13] computes them across autosomes 1–22 and X. Unlike the other three, this one does not read the genealogy at all: it measures the genome, so it *tests* the relationship a shared tail asserts rather than adding to the assertion. A tail implying a common ancestor g generations back predicts an expected range of shared centimorgans; agreement corroborates the paper trail, and disagreement localises an error in it. The check has a range limit, since detectable autosomal sharing falls away beyond roughly six to eight generations, so it validates only the recent end of a path. That is the same window in which, by the argument of Section 10.1, the distinguishing information sits, so the two lines of reasoning bound the useful depth from opposite directions.

Combining the four pieces of evidence, the length of the shared tail, the name sound-alike, haplogroup agreement on the uniparental edges, and autosomal centimorgan sharing, in a

standard record-linkage score [10] gives a match *probability*. We present this as a *proposed* method: the false-positive model of Section 10.1 motivates it analytically, but empirical evaluation on a labeled genealogical corpus (precision and recall against known relationships) remains future work.

10.1 Why it works: far fewer real ancestors than possible paths

About 1.17×10^{11} people have ever lived [19]; since $\log_2(1.17 \times 10^{11}) \approx 37$, roughly 37 yes/no facts (37 bits) are enough to give *every* human who has ever lived a distinct label, close to the smallest code possible. (For scale, 37 generations span about a millennium; reaching an ancestor more distant than that needs a longer path, but pedigree collapse, below, holds the number of *distinct* ancestors far below the naive count of 2^L possible paths.) An `atree` path is also a label relative to its starting person: the same ancestor has different paths from different descendants (and several when collapse occurs), so the 37-bit figure says only that a short label *exists*; it does not claim that an `atree` path is by itself a single global ID for that person, and building such an ID is the collapse-aware merge of Section 13. There are 2^L possible paths of length L, far more than the $\leq 10^{11}$ people who could fill them (at L=100 that is $2^{100} \approx 1.3 \times 10^{30}$ slots). The paths actually used are thinly spread. Two unrelated lines share a tail of k steps by chance with probability about 2^{-k} (taking the sexes along a path as roughly even and independent; about 10^{-9} at k=30), and matching names make a coincidence even less likely; a shared tail of 20–30 letters with matching names is almost certainly a real common ancestor. The paths in use are thinly spread *because they are shared*: pedigree collapse, together with the recent common ancestry of all living people [3, 23], means deep ancestors are common to nearly everyone (the tails converge) while people differ mainly in recent generations. So the information that tells people apart sits in the recent end of the path (the first $\sim$20–35 letters), and the number of distinct lines is about the population, $\sim\log_2$(population) $\approx$ 37 bits, not the naive 100. The huge count of possible paths is an artifact of writing the family graph out as a tree. The real object is a small, shared graph, and identifying a person in it is again a form of compression.

10.2 A scoring model

Record linkage scores a candidate pair by summing the log likelihood ratio of each piece of evidence [10]; in bits, the terms add. Writing R for "the pair is related as the shared tail asserts":

Evidence	$\log_2$ LR (bits)	Where it is valid
Shared tail of k steps	k	recent range only (Sec. 10.1)
Name agreement	$\log_2(m/u)$	m, u estimated from a labelled corpus
Haplogroup concordant, freq. f	$\log_2(1/f)$: 1.0 at f=0.5, 2.3 at f=0.2, 4.3 at f=0.05	all-`M` / all-`F` edges only
Haplogroup discordant	large negative (exclusion)	all-`M` / all-`F` edges only
Autosomal cM sharing	likelihood over g	$g \lesssim$ 6–8 generations

The tail term follows from Section 10.1: two unrelated lines share a k-step tail with probability about 2^{-k}, while related ones share it by construction, so concordance is worth about k bits. This holds only in the range where the independence assumption does: beyond roughly 20–30 generations, universal recent common ancestry [3, 23] makes deep tails common to nearly everyone, and the term must not be read as 2^k there. The model is therefore strongest exactly where Section 10.1 locates the distinguishing information, in the first 20–35 letters.

The haplogroup term is asymmetric, and this is its main practical point. Haplogroups are deep clades, unchanged over genealogical time, so concordance is expected of a true relative but

also of everyone else in the clade: its evidential value is only $\log_2(1/f)$, which for a common haplogroup is one or two bits, an order of magnitude less than a ten-step tail. Discordance is close to decisive: a mismatch along an unbroken patriline or matriline refutes the asserted path. The field is an exclusion test more than a confirmation. It speaks only about the two extreme paths; a mixed `M/F` tail receives no uniparental evidence at all.

The autosomal term is the only one that measures anything outside the record. Expected sharing falls with relationship distance, so an observed centimorgan total induces a likelihood over g that can be compared with the g the tail asserts. Haplogroups cannot supply this, having no genealogical-timescale clock; short tandem repeat haplotypes, which mutate at a rate per marker per generation, could supply a second such estimate along the patriline.

We give the model rather than fitted parameters: m and u, the haplogroup frequencies, and the expected-sharing curve are all quantities the literature or a labelled corpus supplies, and estimating them here without such a corpus would be a fit to nothing. The contribution is the decomposition and its stated validity limits.

11 Related work

The dominant genealogical interchange format is GEDCOM [9] (and the later GEDCOM X [17]): record- and pointer-oriented, and general enough to represent arbitrary family graphs. `atree` complements it: a compact, line-oriented encoding specialized to the *ascending* tree, in which each line is self-sufficient. The difference is in the structure. GEDCOM identifies individuals by cross-reference, so a record's meaning depends on the records it points to, and editing by hand means preserving referential integrity by hand. That design is what makes GEDCOM general, and also what makes a single record unusable in isolation. In `atree` the path is the position, so there is nothing to dereference. When the format was designed (2007–2011), genealogies were kept either in application-specific stores or in GEDCOM, so correcting or exchanging part of a tree meant working through an application; independence from any application was the motivating requirement. The author's own open-source tools realize the surrounding interoperability: `aisgedcom` converts between GEDCOM and `atree` [14]; `aisconvert` [13] parses 23andMe/FTDNA genomes and computes HIR / centimorgan overlaps (with `ped2raw` [16] for PED→RAW conversion); and `nid` [15] produces anonymized identifiers by cryptographic hashing, relevant to the entity-resolution setting. Among genealogical numbering systems, ascendant numbering is dominated by Ahnentafel / Sosa–Stradonitz [1, 8], while descendant lines use the Henry, d'Aboville, de Villiers–Pama, and Register systems [18]; `atree` is the binary (path) form of Ahnentafel and inherits its two-parent structure. Anderson [2] described the path reading earlier and observed that two people's charts can be compared to find their relationship; the attribution section states what `atree` adds. The matching method of Section 10 builds on classical record linkage [10] and phonetic name coding [22, 24], contributing the `atree` path suffix as a compact, name-independent blocking key, complementary to autosomal-segment (HIR / centimorgan) and uniparental (Y/mtDNA) evidence.

12 Prior publication

The `atree` specification, with its BNF grammar, was first published by the author on 7 November 2011 [11]; the corresponding section of the Wikipedia article *Genealogical numbering systems* [18] was added the following day. The reference implementations predate both (`aisconvert`, 2009; `aisgedcom`, 2011). The Wikipedia section was written by the author, so it is disclosed here as provenance and is not cited as independent corroboration. The present paper consolidates and extends that material: the grammar, the bijection, the tooling description, and the lineage-matching analysis of Section 10 appear here for the first time in a single reviewed account. Both

2011 sources are archived by the Internet Archive.[2]

13 Conclusion

`atree` encodes a direct ancestor as the `M/F` path from the subject, the binary form of the Ahnentafel number, giving identifiers that are logarithmic in tree size, self-sufficient, and plain-text (hence diffable, mergeable, and durable). It comes with a grammar, bijective converters, and GEDCOM interoperation. Its main payoff is matching: a shared path suffix, combined with phonetic name agreement and uniparental haplogroup concordance, gives a quantifiable, name-robust signal for genealogical entity resolution, and $\sim$37 bits suffice to address any human who has ever lived, identification as compression, a perspective developed in the author's companion work on intelligence as the discovery of compressors [12]. Open directions: a collapse-aware merge that folds the tree-unfolding back to the underlying DAG, and joint structural/phonetic/genetic scoring at scale.

Acknowledgements and attribution

The bare sex-path notation, a string of `M/F` letters (e.g. `MMFFFMFM`), was already in circulation in genealogy forum discussion, used by example and without a name; the author met it there and thanks the correspondent who introduced it. The notation is pre-existing. The earliest published description the author has located is Anderson's, in print by 1995 [2], which gives the same father/mother reading of the binary Ahnentafel expansion in `f/m` form. The author was unaware of it: Anderson's two-page note appeared in a regional genealogical society journal and did not surface in the general sources consulted at the time, among them the Wikipedia survey of genealogical numbering systems, which then listed no binary or path-based notation at all. The format described here was therefore developed without reference to that precursor, which is a statement about the author's knowledge and not a claim of priority over it. The author's contributions over that precursor are the **name "`atree`"**; the **self-contained plain-text format and grammar**; the **reference implementations**, as sole programmer; the scored **lineage-matching analysis** of Section 10, which combines suffix, phonetic, uniparental and autosomal evidence with stated validity limits; and the structural comparison of two charts, which Anderson also observed, given a grouping key and a score.

The author's grounding in graph theory owes much to the advanced graph-theory course of the late Professor Shimon Even (1935–2004) [6, 7], whose student he was.

Code and data availability

The `atree` reference implementation and related tooling are open source: `aisgedcom` (GEDCOM↔`atree` conversion and `AtreeParser`) [14]; `aisconvert` (HIR / centimorgan calculator) [13]; `ped2raw` [16]; and `nid` [15]. The two reference shell converters are reproduced in full in Appendix B. No new datasets were used; the record in Section 5 is an illustrative excerpt. These tools were written in 2009–2011 and are no longer maintained; they are provided as historical artifacts, for provenance and reference, and may not build or run unmodified on current systems. The self-contained shell converters of Appendix B, by contrast, are current: run under a POSIX shell, `s2atree.sh MFFMMMMFFFMMF` returns `7225` and `atree2s.sh 7225` returns the same path, differing only in the leading subject-sex letter, which Ahnentafel does not encode and which the script takes from its `my_gender` setting (Proposition 1). Conversion was checked at every depth

[2] http://web.archive.org/web/20260607185709/https://siberean.livejournal.com/14874.html and http://web.archive.org/web/20260420170149/https://en.wikipedia.org/wiki/Genealogical_numbering_systems.

from 1 to 200: the path length matches the generation distance, the alphabet is closed, and the round trip through `s2atree.sh` returns the original number exactly.

A A template / example file

Person-field forms (each real line is prefixed by an `atree M/F` path; the field forms below are the part after the path). Surname in `/.../`, maiden name in parentheses, `?` and `//` for unknowns, a date acts as the name/place delimiter, and `#` starts a comment.

```
FirstName
FirstName (1600)
FirstName (ABT 1600)
FirstName (14 AUG 1855-5 APR 1877)
FirstName /Surname/
FirstName /Surname/ (?-ABT 1800)
FirstName /Surname/ (AFT 1800-?)
/Surname/ (28 OCT 1852-BEF 1880)
FirstName /Surname (MaidenSurname)/ (1764-6 OCT 1813)
FirstName Another Name /Surname (MaidenSurname)/ (14 AUG 1855-5 APR 1877), Place of origin
FirstName // (1600)        # nothing but a date is known
//                         # nothing is known
/?/                        # surname refinement is the priority
FirstName /Surname (?)/    # maiden surname unknown
# a comma before the given name, as some languages / GEDCOM require:
/Surname/, AnotherName (1800), Place of origin
# if a place could be confused with a comma in a name, the date is the delimiter:
Names (?) Place of origin
```

B Reference converter scripts

The two reference converters (subject sex set at the top; `bc` does the base conversion).

```
#!/bin/sh
# atree2s.sh : convert an Ahnentafel number to an atree path  (POSIX sh).
# Usage: atree2s.sh <ahnentafel-number>

my_gender=F          # subject's sex: M or F
male=M
female=F

case "$1" in
    '')        echo "usage: $0 <ahnentafel-number>" >&2; exit 1 ;;
    *[!0-9]*)  echo "$0: '$1' is not a number"      >&2; exit 1 ;;
esac
case "$1" in
    *[!0]*)    : ;;
    *)         echo "$0: Ahnentafel numbers start at 1" >&2; exit 1 ;;
esac

# Ahnentafel number -> binary -> letters (0 = father M, 1 = mother F).
# bc folds long output at 70 columns with a backslash-newline;
```

```
# strip the continuations before touching the digits, or paths
# deeper than 67 generations are silently corrupted below.
binary=$(echo "obase=2; $1" | bc | tr -d '\\\n')
letters=$(printf '%s' "$binary" | tr '0' "$male" | tr '1' "$female")

# The leading bit denotes the subject; rewrite it as the subject's sex.
letters=$(printf '%s' "$letters" | sed "s/^./$my_gender/")

echo "$letters"
```

```
#!/bin/sh
# s2atree.sh : convert an atree path to its Ahnentafel number  (POSIX sh).
# Usage: s2atree.sh <atree-path>

male=M
female=F

case "$1" in
    '')        echo "usage: $0 <atree-path>"        >&2; exit 1 ;;
    *[!MF]*)   echo "$0: '$1' is not an atree path" >&2; exit 1 ;;
esac

# The first letter is the subject; the Ahnentafel leading bit is always 1,
# so normalise it to F before mapping letters back to bits (M = 0, F = 1).
path="$female$(printf '%s' "$1" | cut -c2-)"
binary=$(printf '%s' "$path" | tr "$male" '0' | tr "$female" '1')

echo "ibase=2; $binary" | bc
```

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